

Comparative Analysis of Machine Learning Classifiers for Multiclass Classification Problem

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Abstract

Multiclass classification is a fundamental problem in machine learning where the instances must be categorized into more than two classes. This study presents a comparative analysis of several multiclass classifiers applied on 5 diverse datasets such as network strength and speed, retail analytics, etc. Classifiers used techniques like OvO (One vs One), OvR (One vs Rest), ECOC (Error-Correcting Output Codes), Classification DAGs using various models. Classifiers which are inherently multiclass like Decision Tree, SVMs are also used.

1. Introduction

Multiclass classification is a fundamental problem in machine learning in which each instance is assigned to one of three or more predefined classes. Unlike binary classification, which involves only two possible outcomes, multiclass problems require models to distinguish among multiple categories simultaneously, making them inherently more complex. This increased complexity arises from overlapping class boundaries, higher dimensional decision spaces, and the need for scalable learning algorithms.

Multiclass classification has a wide range of real-world applications across various domains. In wireless communication, it is used for predicting signal strength and optimizing network coverage. In retail analytics, it enables customer segmentation and behavior prediction, supporting targeted marketing strategies. In computer vision, datasets such as Fashion MNIST serve as benchmarks for evaluating classification models on image data. Similarly, in network performance analysis, multiclass models are used to categorize internet speed or quality of service levels. These diverse applications highlight the importance of robust and adaptable classification techniques.

Despite its significance, multiclass classification presents several challenges. Many traditional machine learning algorithms, such as Support Vector Machines and Logistic Regression, are inherently designed for binary classification and must be extended to handle multiple classes. Additionally, issues such as class imbalance, noisy data, and varying dataset structures can significantly impact model performance. The choice of an appropriate classification strategy therefore becomes crucial.

To address these challenges, several decomposition-based techniques have been proposed, including One-vs-Rest (OvR), One-vs-One (OvO), and Error-Correcting Output

Codes (ECOC). These approaches transform multiclass problems into multiple binary classification tasks, allowing the use of well-established binary classifiers. Alongside these methods, certain models such as Decision Trees and Support Vector Machines (in specific configurations) can inherently handle multiclass problems.

In this study, a comprehensive comparative analysis of multiple multiclass classification techniques is conducted across five datasets from diverse domains. Various classifiers combined with different multiclass strategies are evaluated using consistent preprocessing, validation, and performance metrics. The primary objective is to analyze how different models and strategies perform under varying data characteristics and to identify patterns that can guide the selection of suitable approaches for multiclass classification tasks.

2. Literature Review

Multiclass classification is an essential task in machine learning that involves assigning instances to one of multiple predefined categories. Unlike binary classification, where only two classes are considered, multiclass problems require models to distinguish among several possible outcomes, making them inherently more complex. Applications of multiclass classification are widespread, including image recognition, customer segmentation, medical diagnosis, and wireless signal prediction.

Despite its importance, multiclass classification presents several challenges. These include class imbalance, overlapping class boundaries, and increased computational complexity when scaling to large datasets. Additionally, many traditional classification algorithms are inherently binary and require adaptation to handle multiple classes effectively.

To address these challenges, several strategies have been developed, including One-vs-Rest (OvR), One-vs-One (OvO), and Error-Correcting Output Codes (ECOC). These approaches decompose multiclass problems into multiple binary classification tasks, enabling the use of standard classifiers such as Logistic Regression and Support Vector Machines.

This study aims to perform a comprehensive comparison of various multiclass classification techniques across five datasets from different domains. By evaluating multiple models and strategies under consistent conditions, this work seeks to identify trends in performance and provide insights into the suitability of different approaches for diverse data scenarios.

3. Methodology

3.1. Datasets

The study uses 5 datasets from diverse domain applications to ensure a comprehensive evaluation of multiclass classification techniques.

1. *Mobile Communication System Measurements and Satellite Images* [2]:

The dataset consists of fixed sized patch satellite images of a Danish university campus along with positional data such as coordinates and base transmitters. The dataset was published with the paper in reference [1]. The task is to predict path loss at each position. Initially a regression task, it is converted to multiclass classification.

2. *Consumer Shopping Trends Analysis* [3]:

This is a collection of user data such as demographics, behaviour and purchasing habits of customers like store visits, spendings, brand loyalty, etc. The task is to predict their shopping preferences.

3. *Online Retail Customer Segmentation* [4]:

Contains structured customer behaviour data from online retail environments. It is designed for supervised multiclass classification tasks involving customer tier prediction.

4. *Fashion MNIST* [5]:

A simple MNIST like dataset of Zalando's fashion article images. It contains 28x28 greyscale images. The task is to predict the fashion accessory that the image depicts.

5. *Internet Speed Prediction* [6]:

The dataset consists of network data such as latency, download speed, router distance, etc to predict the network strength. Originally designed for a regression task, it is converted to a multiclass classification problem.

3.2. Preprocessing

Prior to evaluation, all data was manually pre-processed. All rows with missing values were removed. Suitable encoding techniques were employed such as categorical (one-hot) and ordinal encoding for variables. Taking in account the limited computational power, subsets of the chosen datasets are used.

Dataset 1 and Dataset 5 were initially designed for regression tasks. Their target variables are discretized to convert the problem into a multiclass one. Discretization for Dataset 1 was performed using a threshold, whereas for dataset 5 it was done using equal frequency Quantile Binning. For Dataset 1, the target variable signal strength is discretized into 3 classes: Strong, Moderate and Poor. Signal strength less than -100 dBm was categorized as 'Poor', between -100 to -80 as 'Moderate' and above -80 as 'strong'. For Dataset 5, 4 equal frequency divisions were made for the target variable.

The following table contains information about the features, classes and instances for each dataset. All except dataset 1 contains an equal number of instances per class. Instances per class for dataset 1 are also mentioned in the table below:

S.no	#Instances	#Features	#Classes	#Instances per Feature
1	800	2058	3	320 (0), 332 (1), 148 (2)
2	1107	26	3	369
3	8000	14	4	2000
4	4000	785	10	400
5	5000	13	4	1250

3.3. Classification Models

Inherently multiclass algorithms like Decision Trees, SVMs are used while Logistic Regression model with OvO, OvA, ECOC strategy are used. An implementation of Classification DAGs is also used.

In the **One-vs-Rest (OvR)** approach, a separate classifier is trained for each class against all other classes. This method is computationally efficient and widely used, particularly for large datasets. However, it may be sensitive to class imbalance, as each classifier is trained with uneven class distributions.

The **One-vs-One (OvO)** strategy constructs a binary classifier for every pair of classes. This results in a larger number of classifiers but often leads to improved decision boundaries, especially in datasets with complex class separations. Final predictions are typically made using a voting mechanism.

The **Error-Correcting Output Codes (ECOC)** method represents each class using a unique binary code and trains multiple classifiers based on this encoding. This approach introduces redundancy, allowing the system to correct certain classification errors and improve robustness, particularly in noisy datasets.

In addition to these decomposition techniques, a **Directed Acyclic Graph (DAG)-based classifier** was also employed. This method organizes multiple binary classifiers in a graph structure, enabling efficient decision-making by progressively eliminating classes during prediction.

Furthermore, classifiers that inherently support multiclass classification were also utilized. Decision Trees naturally handle multiclass outputs without requiring decomposition. Similarly, Support Vector Machines were implemented using a multiclass configuration based on the OvO strategy.

3.4. Evaluation Metrics

Model performance was evaluated using multiple metrics including Precision, Recall, Accuracy and F1-Score with both macro and micro averaging. These metrics provide a holistic and balanced explanation for datasets with multiple classes. The metrics used are:

Accuracy: It is the ratio of correctly classified instances to the total number of instances.

Precision: It is the proportion of correctly predicted instances among all instances predicted for a given class. It reflects the model's ability to avoid false positives.

Recall: (or Sensitivity) measures the proportion of actual instances of a class that are correctly identified by the model.

F1-score: It is the harmonic mean of precision and recall and provides a balanced measure of model performance. It is particularly useful when there is an uneven class distribution.

3.5. Validation Strategy

To ensure a robust performance estimate, stratified 5-Fold Cross Validation was used. The approach reduces variance and ensures that all data samples contribute to both training and test phases.

4. Results

The table below shows the results of machine learning algorithms as tested on the chosen datasets.

Results for Dataset 1 (Network Strength):

Model	Accuracy	Recall	Precision	F1
OVR (LR)	0.78±0.03	0.78±0.03	0.78±0.03	0.76±0.04
OVO (LR)	0.81±0.03	0.81±0.03	0.81±0.03	0.8±0.04
ECOC (LR)	0.73±0.04	0.73±0.04	0.73±0.04	0.67±0.09
DT	0.77±0.04	0.77±0.04	0.77±0.04	0.77±0.05
SVM	0.47±0.02	0.47±0.02	0.47±0.02	0.33±0.02
DAG	0.74±0.06	0.74±0.06	0.74±0.06	0.73±0.07

Results for dataset 2 (Consumer Shopping Trend Analysis):

Model	Accuracy	Recall	Precision	F1
OVR (LR)	0.95±0.01	0.95±0.01	0.95±0.01	0.95±0.01
OVO (LR)	0.94±0.01	0.94±0.01	0.94±0.01	0.94±0.02
ECOC (LR)	0.71±0.21	0.71±0.21	0.71±0.21	0.69±0.24
DT	0.74±0.02	0.74±0.02	0.74±0.02	0.74±0.02
SVM	0.72±0.03	0.72±0.03	0.72±0.03	0.72±0.02
DAG	0.74±0.03	0.74±0.03	0.74±0.03	0.74±0.03

Results for dataset 3 (Retail Customer Segmentation):

Model	Accuracy	Recall	Precision	F1
OVR (LR)	0.61±0.02	0.61±0.02	0.61±0.02	0.6±0.01
OVO (LR)	0.62±0.02	0.62±0.02	0.62±0.02	0.62±0.02
ECOC (LR)	0.49±0.1	0.49±0.1	0.49±0.1	0.49±0.11
DT	0.65±0.01	0.65±0.01	0.65±0.01	0.65±0.01
SVM	0.24±0.01	0.24±0.01	0.24±0.01	0.17±0.01
DAG	0.65±0.01	0.65±0.01	0.65±0.01	0.64±0.01

Results for dataset 4 (Fashion MNIST):

Model	Accuracy	Recall	Precision	F1
OVR (LR)	0.76±0.01	0.76±0.01	0.76±0.01	0.76±0.01
OVO (LR)	0.81±0.01	0.81±0.01	0.81±0.01	0.81±0.01
ECOC (LR)	0.7±0.02	0.7±0.02	0.7±0.02	0.7±0.02
DT	0.73±0.01	0.73±0.01	0.73±0.01	0.73±0.01
SVM	0.83±0.01	0.83±0.01	0.83±0.01	0.83±0.01
DAG	0.73±0.01	0.73±0.01	0.73±0.01	0.73±0.01

Results for dataset 5 (Internet Speed Prediction):

Model	Accuracy	Recall	Precision	F1
OVR (LR)	0.94±0.02	0.94±0.02	0.94±0.02	0.94±0.02
OVO (LR)	0.98±0.0	0.98±0.0	0.98±0.0	0.98±0.0
ECOC (LR)	0.61±0.18	0.61±0.18	0.61±0.18	0.53±0.22
DT	0.97±0.0	0.97±0.0	0.97±0.0	0.97±0.0
SVM	0.98±0.0	0.98±0.0	0.98±0.0	0.98±0.0
DAG	0.97±0.0	0.97±0.0	0.97±0.0	0.97±0.0

We can observe that LR with OVR/OVO strategy achieves the best result in most of the datasets. SVM also show similar performance to that of LR in some of the datasets, noticeably 5th and 4th. Dataset 3 was the only one where decision trees performed best. Even so, LR was not far behind. ECOC models performed worse than OVR/OVO for all datasets. DAGs showed variable performance across the dataset. We tested OVR/OVO with DTs and SVMs however the results were similar to that of using them independently. However, Decision Trees showed comparable performance to LR when evaluating F1-Scores.

5. Discussions and Conclusions

We successfully tested various strategies and algorithms for multiclass classification. We systematically reviewed all methods and tested them across diverse datasets such as retail customer data, network strength loss prediction and fashion article image data. We could observe that simple model such as Logistic Regression outperformed models like Decision Trees and SVMs with OVR/OVO strategy. Classification DAGs showed variable results while ECOC performed comparatively poor consistently. The future work could explore more such techniques for further improvement in results in multiclass classification.

6. References

- [1] <https://ieeexplore.ieee.org/document/8950164/algorithms?tabFilter=dataset>)
- [2] <https://ieee-dataport.org/open-access/mobile-communication-system-measurements-and-satellite-images>
- [3] <https://www.kaggle.com/datasets/minahilfatima12328/consumer-shopping-trends-analysis>
- [4] <https://www.kaggle.com/datasets/rohit8527kmr7518/online-retail-customer-classification-dataset>
- [5] <https://www.kaggle.com/datasets/zalando-research/fashionmnist>
- [6] <https://www.kaggle.com/datasets/getanmolgupta01/internet-speed?resource=download>